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Retrofit Thru Tubing Deep Gas Lift System Successfully Enhances Hydrocarbon Production by More Than 200% in Offshore East Java

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Intermittent production and underperforming wells are often encountered due to the dynamic nature of oil and gas reservoirs. Maximizing the potential and reactivation of these wells then become a major priority due to their significant hydrocarbon reserves. To address this efficiently, emphasis must be placed on cost-effective, rigless intervention solutions that minimize operational downtime and optimize return on investment.

This offshore field lies approximately 15 kilometers north of Gresik, East Java, in shallow waters with depths between 10 and 15 meters. Development began in 2017, and production has been ongoing since 2021 through the installation of two wellhead platforms. Because the reservoir is characterized by low permeability and primarily produces oil, pressure depletion has a detrimental effect on the well's ability to lift fluids once below its initial hydrostatic conditions.

Conventional gas lift techniques are commonly used during initial well development for extending its end-of-life production and maximizing recovery. However, as reservoir pressure declines over time, the fluid level may drop below the completion gas lift mandrels, reducing lift efficiency. Inadequate gas lift and reservoir pressure to kick start production may cause wells to shut in and have intermittent production requiring interventions in each downtime which is uneconomical (Brown, 1984). Gas lift optimization method is key in unlocking the most potential out of underperforming wells. Ultimately, the success of such methods depends on maximizing gain while overcoming limitations related to operational cost, accessibility, and logistical challenges.

Introduction

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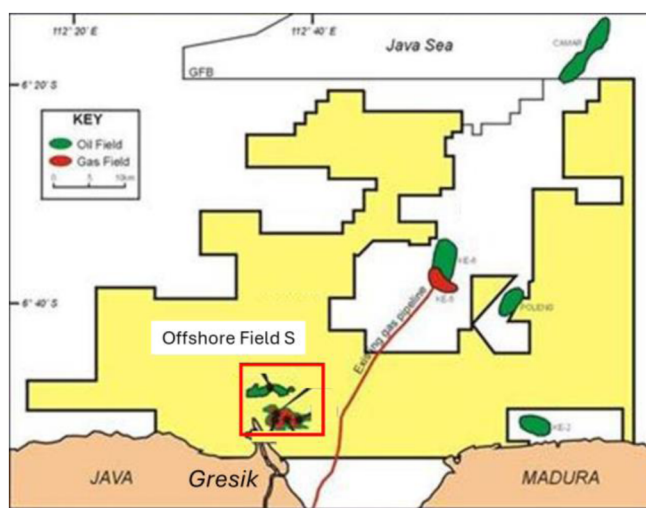


Figure 1—Offshore field S location

Well Information

The relatively new Well S1 was drilled and completed in 2021 with open hole and cased hole multizone 4.5" Tubing completion reaching depth of 7900 ftMD/4813 ftTVDSS. The production zones are accessed through a horizontal section beginning from 6000 ftMD just after the production packer with unopened upside potential on the upper SSDs. Due to the deviation and producing zones of the well, the deepest gas lift mandrel (GLM) available is only at 4466 ftMD deep with 3 additional unloaders above. Meaning that there is significant vertical area of the well that is not supported by lift gas injection.

The well's initial production rate may reach up to 1,200 BOPD during the first month, followed by a decline to approximately 300–400 BOPD as shown in Figure 2. Multiple acidizing operations were attempted to enhance the well permeability and increase the well back to initial potential, however the improvements made couldn't be sustained. Prior to the intervention, a lowering trend is also observed on the production rates with the latest data showing the lowest of 200 BOPD. It was then decided to explore options on major intervention for a more sustainable gain.

The option through a workover is limited due to the presence of upper perforations, which prevent further lowering of the production packer. Furthermore, a workover at this location may not be economically justified given the well's relatively recent completion and the logistical challenges, including limited available space. Therefore, to maximize the well's productivity under these constraints, a cost-effective rigless intervention approach is preferred. Well intervention operation effectiveness will be the key as several oilfields on the block share the same intervention resources. With the high potential of the well, gas lift optimization is the priority on the planning stage of the operation. Several retrofit gas lift options

were explored to select the most effective method, considering operational feasibility, gas lift optimization and investment returns.

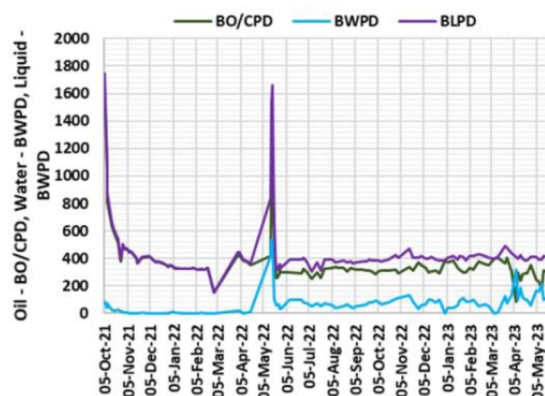


Figure 2—Well S1 production rate graph prior intervention

Siphon string is commonly used to retrofit and deepen the gas lift injection point by directing lift gas from just above the production packer, utilizing a long continuous tubing as a spacer between the straddle packers. Although jointed pipes for the siphon string are effective for gas lift optimization, limited platform access makes deploying a workover unit challenging. High deviation of the well causes slickline intervention to be ineffective on reaching the depth necessary for gas lift deepening. While straddling with coiled tubing as the siphon string can offer operational advantages, it significantly reduces the production cross-sectional area, significantly restricting flow particularly at high rates (Moore, Laflin, & Walker, 1993). An alternative approach is to direct gas lift through coiled tubing from the surface, but this configuration compromises the functionality of the TRSV, making it unsuitable in offshore applications. Therefore, an innovative rigless intervention approach using thru tubing deep gas lift technology was explored to maximize this well potential while overcoming the issues mentioned beforehand. A Pugh matrix is employed to summarize the selection of the methods as presented in Table 1.

Table 1—Retrofit Gas Lift Methods Selection Matrix

Criteria	Weight	Workover	Siphon String Method		Gas Lift Packoff Straddle	Deep Gas Lift using Straddle Packer with CT gas lift deepening
			Jointed Pipe Straddle	Coiled Tubing Straddle		
Conveyance		Workover Unit/Rig	Hydraulic Workover	Coiled Tubing	Slickline	Coiled Tubing
Accessibility	3	-1	-1	1	1	1
Duration	1	-1	0	1	1	1
Gas Lift Depth	3	1	1	1	-1	1
Production flow Area	2	1	1	-1	1	0
Cost	3	-1	-1	0	1	0
Total		-2	-1	5	6	7

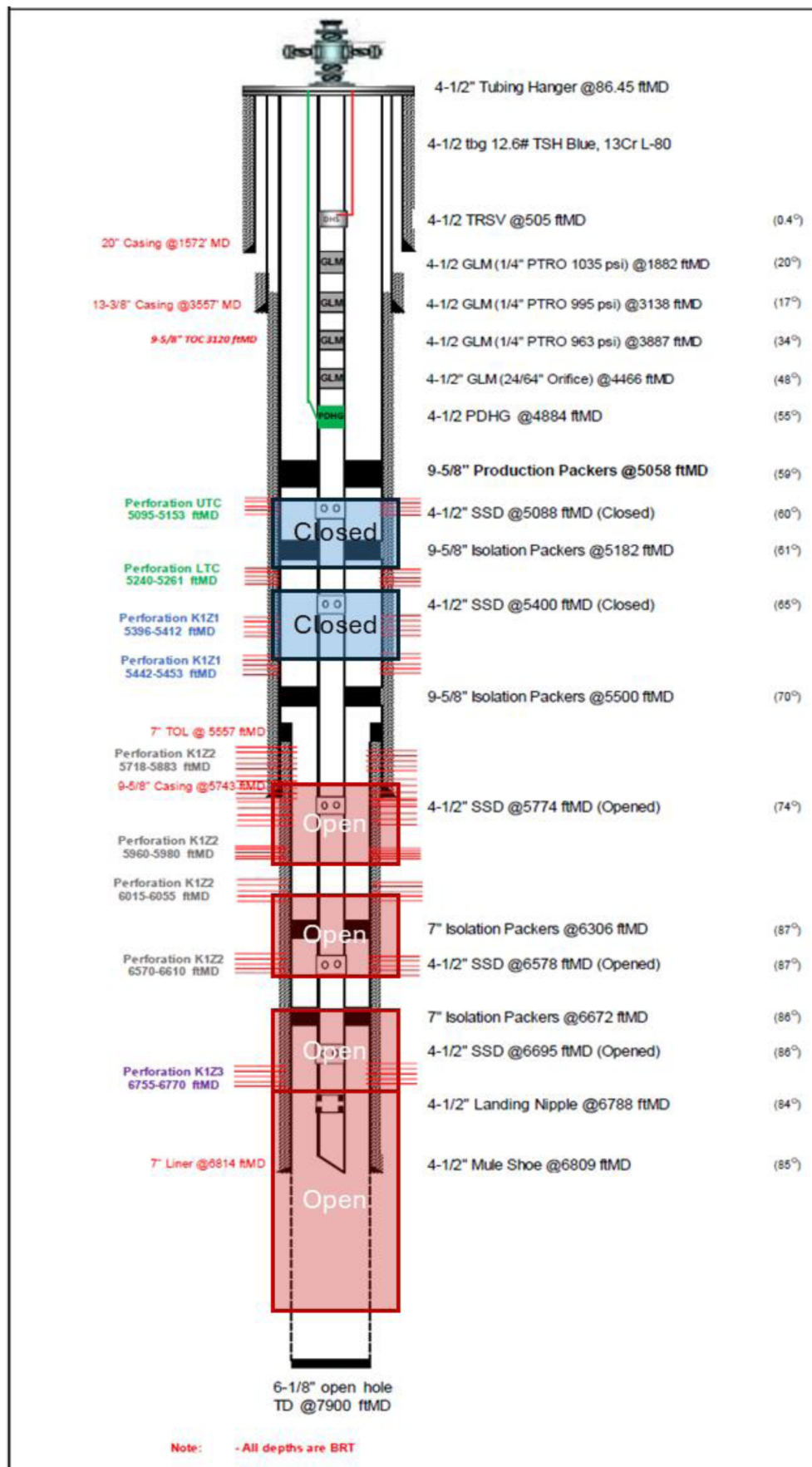


Figure 3—Well S1 Completion Schematic

Deep Gas Lift System Design Concept

The well has an existing gas lift system with the deepest GLM at 4466 ftMD. To utilize the GLM, a seal bore retrievable packer system that straddles the GLM is selected. The system flow path concept is presented in Figure 4. This straddle packer system directs the lift gas from the deepest GLM through a crossflow diverter sub and gas injection tube stinger that regulates the flow path of lift gas and hydrocarbon. The packers only need to straddle the GLM while a coiled tubing string is cut and installed at the tailpipe to deepen the gas injection.

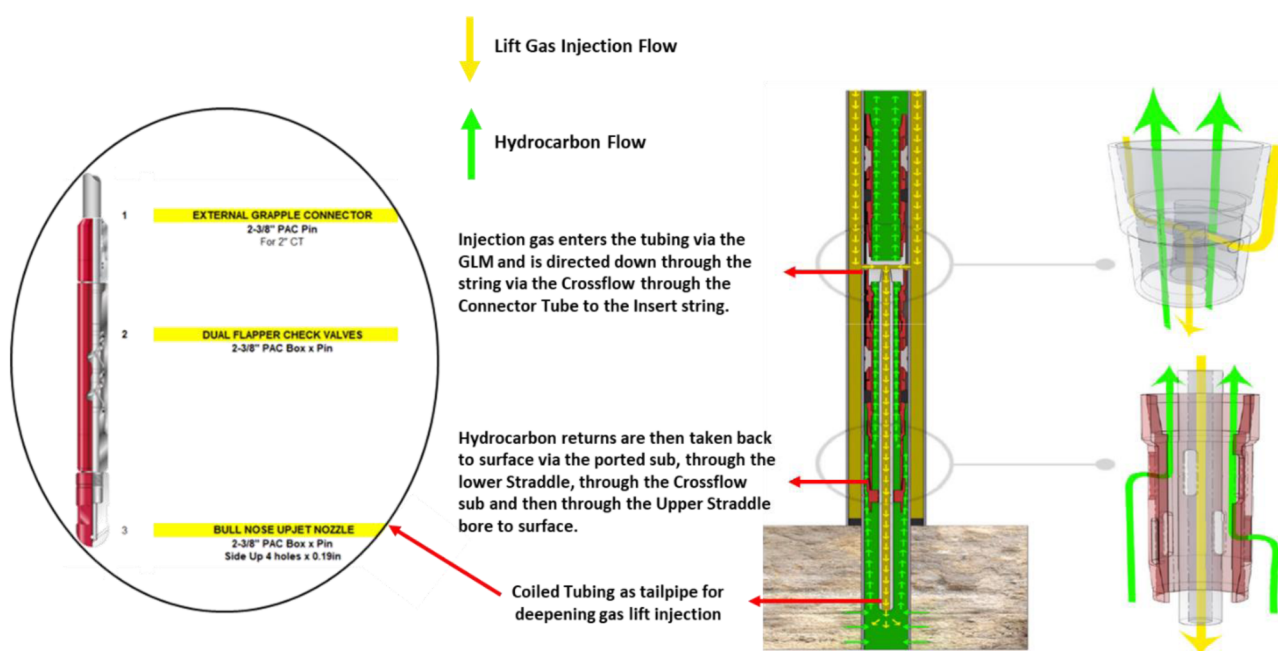


Figure 4—Schematic of the deepening gas lift system flow path

The packers selected for the gas lift deepening system feature gas tight validation according to ISO 14310 V0. This is achieved by bottom hole pressure acting against piston area on the atmospheric chambers that apply and maintain additional compressive force towards the packer element. The V0 rating ensures the packer's reliability in maintaining a gas-tight seal under axial loads of the coiled tubing injection and temperature cycling, providing long-term assurance of integrity. A flapper check valve is installed with an upward jetting nozzle at the injection depth acting as the gas lift valve. Single injection point is chosen for the installation to simplify the operation because the gas lift pressure is considered enough to unload the well directly as explained in the modelling section.

Nodal Analysis

Prior to installation, a production model using nodal analysis is built with the latest available data to simulate the well performance under various conditions. To evaluate performance improvements, two scenarios were modeled: the current system and one with a deep gas lift installation. Furthermore, optimization is done for several parameters of the gas lift, mainly the injection depth, gas lift rate, etc. using sensitivities to look at the feasibility and finalize the design of the deepening gas lift system.

Base Model – Current Conditions

Well S1 is an oil producer which is completed with 4-1/2" 12.6# tubing and produces through two reservoir sections. The existing base model is built with data shown on Table 2. The Flow correlation selected was Hagedorn and Brown (modified) with Well & Riser L-factor tuned to 1.0. Initially, the model calculated

Bottom Hole Pressure (BHP) was slightly different which shows the IPR model differs compared to well test data. Fine tuning on the reservoir pressure, reservoir thickness, permeability and Vogel coefficient was done to match the well-test data. The tuning process with the well test data is plotted in Figure 5.

Table 2—Well modelling input data

Model Input Data			Value	Units
Correlation		Flow Correlation	Hagedorn and Brown (Modified)	
		Liquid Loading	Turner	
Fluid Properties		Fluid Type	Black Oil	
		Layers	2	
		Oil API Gravity	34.7008	deg API
		Gas Specific Gravity	0.75	
Reservoir Properties	Layer 1	Reservoir Pressure	831.92	psig
		Temperature	187.7	deg F
		Depth	6164	ft
		Water Cut	20.1	%
	Layer 2	Reservoir Pressure	787.44	psig
		Temperature	187.7	deg F
		Depth	6755	ft
		Water Cut	20.1	%
Gas Lift Properties		Injection Pressure	1185.3	psig
		Gas Rate	3.1	MMSCF/d

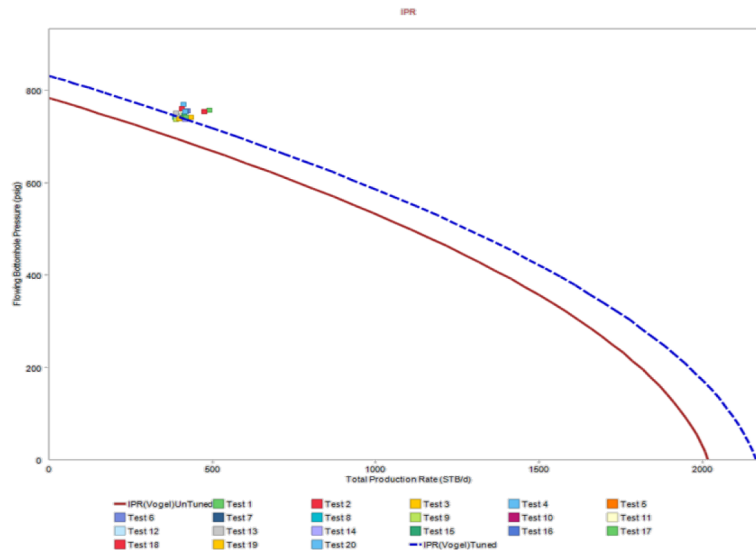


Figure 5—Untuned vs Tuned base model IPR curve

Base Case - Current Conditions Results

The flow curve for current operating conditions based on the existing completion with current gas lift design and lowest valve installed at 4466 ft MD is shown in Figure 6. Tubing head pressure in the well is kept around 260 psi as minimum to be above the flowline pressure and maximum gas lift rate of 3.1 MMSCF/d.

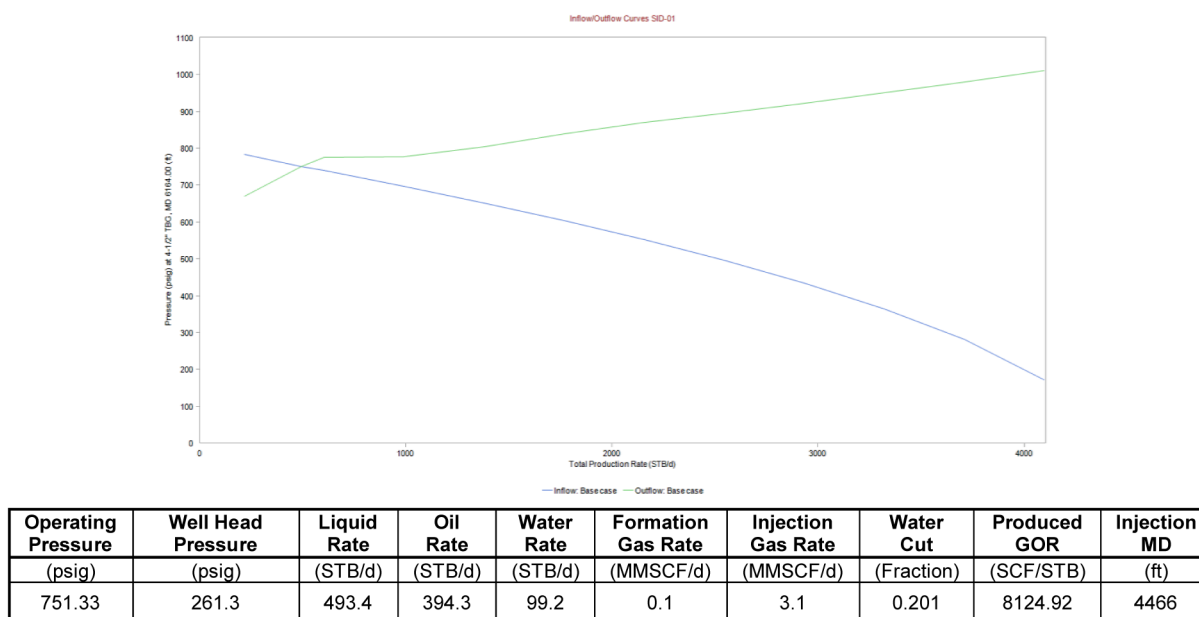


Figure 6—Flow Curves – Current Conditions

Critical unloading velocity for the system is plotted and compared with in-situ gas velocities in Figure 7, where the gas superficial velocities are lower than the critical unloading velocities. Hence, it is confirming that liquid loading is happening in the well. A liquid-loaded well will eventually cease production as the rising liquid level increases hydrostatic pressure beyond the ability of both gas lift and reservoir pressure to sustain flow (Turner, Hubbard, & Dukler, 1969).

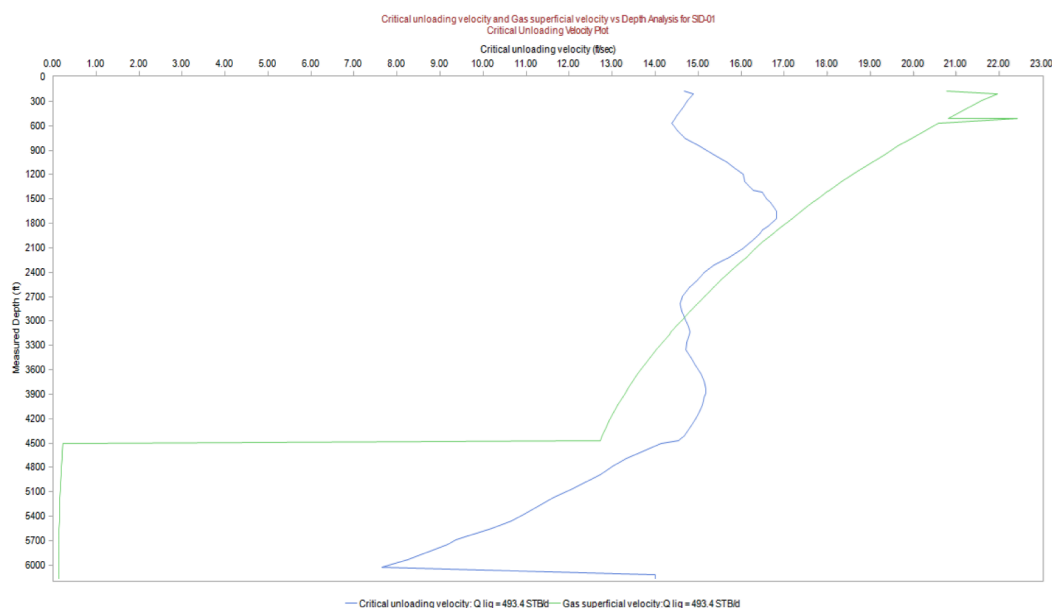


Figure 7—Critical unloading velocity plot – current conditions

Base Case – Sensitivity Analysis; Reservoir Pressures

A sensitivity analysis was run by reducing the reservoir pressures until there was no operating point. This is done to evaluate the current condition well performance in the future where the reservoir pressure declines inevitably. It can be seen in Table 3 that a decline in reservoir pressure had a significant impact on production. When the reservoir pressure drops to 600 psig, the well will not be producing effectively and will be shut

down shortly unless the WHP is reduced or increases the gas injection rate. However, the WHP is already at its minimum considering the flowline of the system. This result shows that the life of the well is not sustainable enough to produce economically and will have a relatively short life.

Table 3—Production summary base case – Sensitivities on reservoir pressure

Sensitivity: Reservoir pressure	Well Head Pressure	Liquid Rate	Oil Rate	Water Rate	Formation Gas Rate	Injection Gas Rate	Injection MD	Status
(psig)	(psig)	(STB/d)	(STB/d)	(STB/d)	(MMSCF/d)	(MMSCF/d)	(ft)	
500								No operating point
600	261.3	2.8	2.2	0.6	0	3.1	4466	Stable
700	261.3	230.7	184.3	46.4	0.05	3.1	4466	Stable
800	261.3	480.5	383.9	96.6	0.11	3.1	4466	Stable

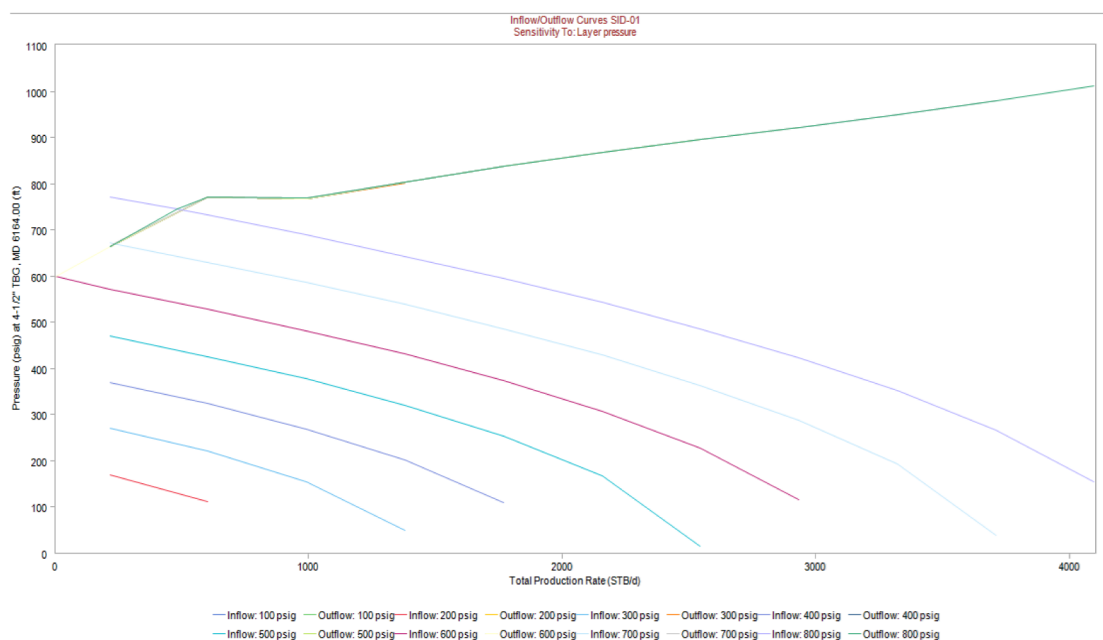


Figure 8—Flow curves base case – Sensitivities on reservoir pressure

Case 2 – Deepening Gas Lift System Model

The tuned base model is then modified to accommodate the deepening gas lift system model as explained in the previous section. As shown in Figure 4, lift gas flows through tubing annulus same as conventional gas lift to the GLM #4 at 4466 ftMD, then continues into the straddle packer system into the injection pipe and coiled tubing tailpipe. This design creates different flow type below GLM #4 where the hydrocarbon flows through annulus between coiled tubing and tubing which is opposite compared to the tubing flow type before GLM. Therefore, casing and tubing are required to interchange based on the flowing area below GLM#4 as illustrated in Figure 9. A minimum flow area of 2.5 in² is also inserted on the model to fully represent the straddle packer system flow restriction. In this operation, 2" coiled tubing with 0.145" thickness is chosen as the injection pipe due to availability.

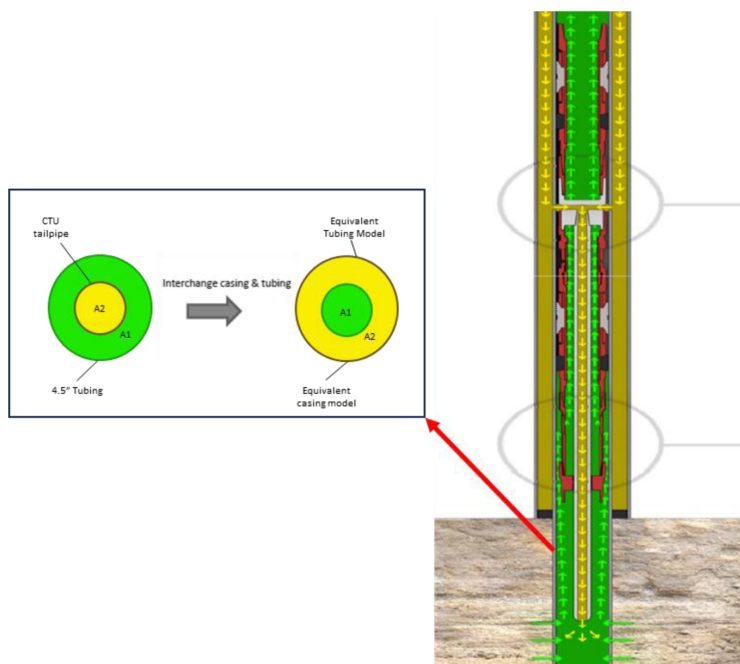


Figure 9—Equivalent casing and tubing area on coiled tubing injection below GLM

Case 2 – Deepening Gas Lift – Injection Depth Analysis

The deepening gas lift system will be able to put injection gas lift as close to the reservoir as possible without operational difficulty. However, the optimized depth needed to be determined first due to gas lift surface equipment limitations. Injecting as deep as possible doesn't necessarily increase production if the available gas lift pressure is insufficient downhole (Yadua, et al., 2021). Injection depth is analyzed by comparing pressures between gas lift flow path (casing side) and hydrocarbon (tubing side) with the deepening gas lift system installed. Figure 10 shows the plotted pressures vs depth on the tubing (production) and casing (gas lift) with gas lift surface pressure of 1185.3 psi and rate of 3.1 MMSCF/d.

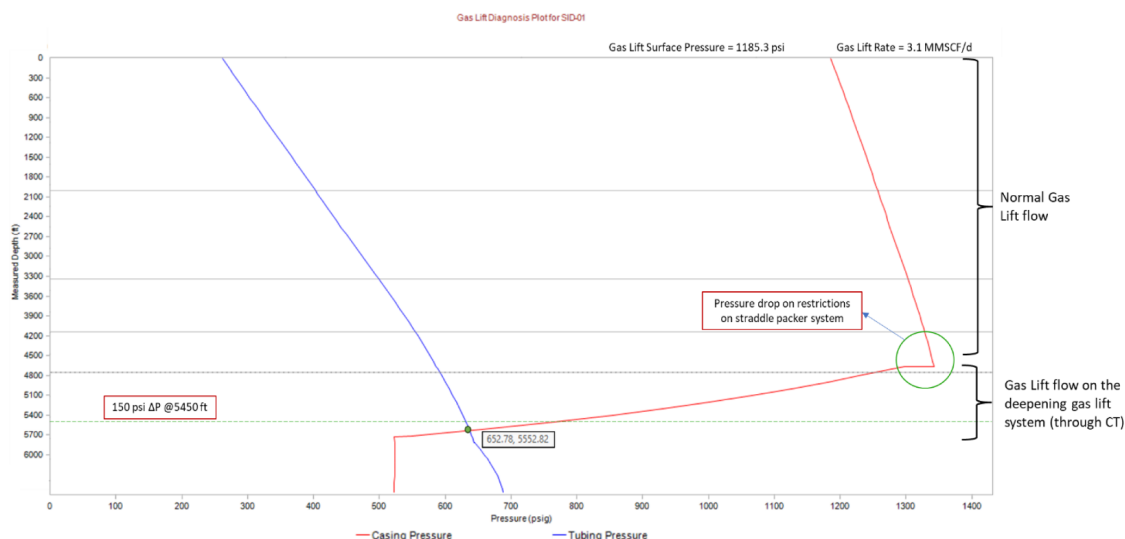


Figure 10—Production flow path (equivalent tubing) and gas lift flow path (equivalent casing) pressure vs depth graph

The gas lift flow area from surface up to GLM#4 is the annulus between 9-5/8" casing and 4-1/2" tubing, which results in low friction loss. A gradual increase in gas lift pressure is still observed with increasing depth, up to GLM#4 where the straddle packer system is installed. Starting from GLM#4, significant friction

loss occur on the gas lift caused by flowing through the restricted area of the coiled tubing. With 50 psi as the required differential pressure between annulus and tubing, 5500 ftMD is determined to be the deepest possible injection point with the chosen deepening gas lift system.

Sensitivity analysis on the resulting produced liquid rate is also conducted by varying the injection depth between 4,800 ftMD and 5,466 ftMD. The results are plotted in Figure 11 as a correlation between injection depth and its resulting liquid production rate. A linear trend shows an increase of liquid rate the deeper the injection depth. It can be concluded that going with the deepest possible injection depth is still worth the gain. Therefore, with safety factors considered, an injection depth of 5466 ftMD is chosen for the end of the deepening gas lift system which is 1000 ft of deepening from the initial GLM depth. Given the well trajectory, where the horizontal section begins at approximately 6,000 ft, further deepening offers minimal benefit, as the impact on hydrostatic pressure becomes negligible.

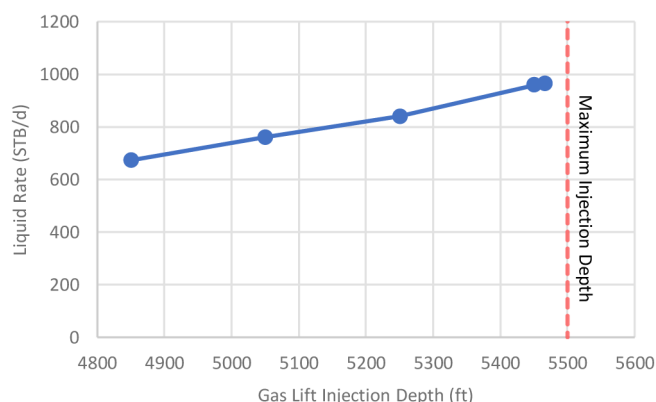
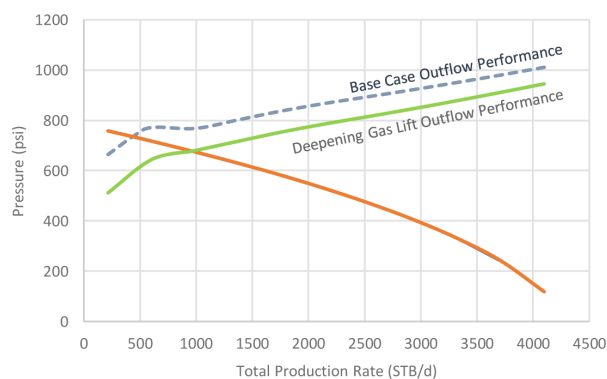


Figure 11—Sensitivities with gas lift injection depth with resulting produced liquid rate

Case 2 – Deepening Gas Lift Results

The flow curve for the retrofit deepening gas lift system with injection depth of 5466 ft MD is shown in Figure 12. The curve is compared to the base condition curve to see the effect of deepening gas lift injection point to the outflow curve. The deeper gas injection effect is seen as a shift towards producing at a lower BHP. The well is still relatively vertical until 6000 ftMD, the result shows that gas lift access to reduce 1000 ft more liquid column significantly affects the well operating pressure up to 100 psi.



Operating Pressure	Tubing Head Pressure	Liquid Rate	Oil Rate	Water Rate	Formation Gas Rate	Injection Gas Rate	Water Cut	Produced GOR	Injection MD
(psig)	(psig)	(STB/d)	(STB/d)	(STB/d)	(MMSCF/d)	(MMSCF/d)	(Fraction)	(SCF/STB)	(ft)
677.34	261.3	964.1	770.3	193.8	0.22	3.1	0.201	4309.22	5466

Figure 12—Flow curves of base case conditions vs retrofit with deepening gas lift system

Critical unloading velocity for the retrofitted system is plotted and compared with in-situ gas velocities in Figure 13. Comparing the gas velocities with base scenario, there are significant portions where the deeper gas lift injection increases the velocity above critical unloading velocity. Below the gas lift injection point, gas velocity is still lower than the critical velocity. These results suggest that the well may still experience liquid loading, although at a slower rate than previously condition.

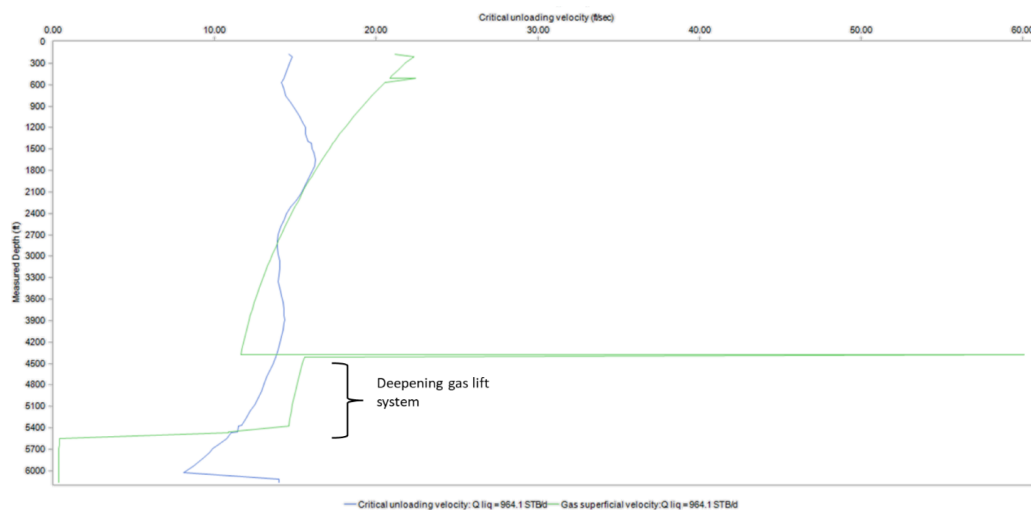


Figure 13—Critical unloading velocity plot – Deepening Gas Lift installed

Case 2 – Deepening Gas Lift – Sensitivities on Injection Gas Rate

Figure 14 and Table 4 shows the production results obtained after installing retrofit deepening gas lift system while injecting gas at different rates from 1.5 to 3.5 MMSCF/d. Produced liquid rates suggest that an optimum gas injection rate of around 2.5-3 MMSCF/d, as the table shows no further improvement above this rate. However, as shown in Figure 13, in-situ gas velocities are higher than critical velocity with a small margin with the surface equipment maximum injection rate of 3.1 MMSCF/d. Therefore, a gas injection rate of 3.1 MMSCF/d was selected to achieve a balance between maintaining critical velocity and optimizing production rate.

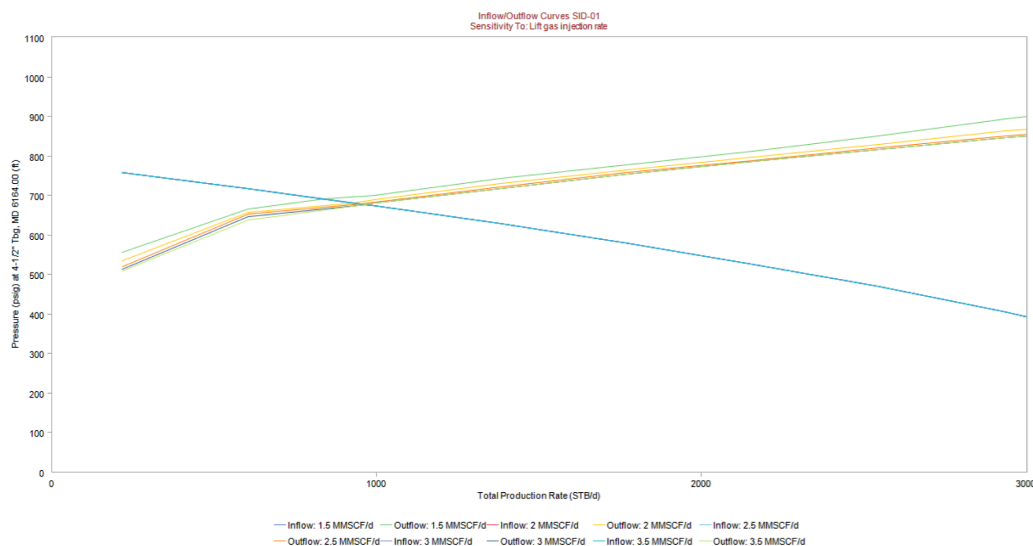


Figure 14—Flow curves retrofit gas lift – Sensitivities on injection gas rate

Table 4—Production summary retrofit gas lift – Sensitivities on injection gas rate

Operating Pressure	Liquid Rate	Oil Rate	Water Rate	Formation Gas Rate	Water Cut	GOR	Injection MD	Lift gas injection rate
(psig)	(STB/d)	(STB/d)	(STB/d)	(MMSCF/d)	(Fraction)	(SCF/STB)	(ft)	(MMSCF/d)
691.48	838.6	670.1	168.6	0.19	0.201	2523	5466	1.5
681.67	926	739.9	186.1	0.21	0.201	2987.79	5466	2
678.6	953	761.5	191.6	0.22	0.201	3567.88	5466	2.5
677.4	963.5	769.9	193.7	0.22	0.201	4181.54	5466	3
677.6	961.8	768.5	193.3	0.22	0.201	4839.12	5466	3.5

Case 3 – Deepening Gas Lift – Sensitivities on Reservoir Pressure

To compare the effect of declining reservoir pressure with the base case, a sensitivity analysis was run by reducing the reservoir pressures until there was no operating point. Figure 15 and Table 5 shows the effect of reducing the reservoir pressure on production rates. When the reservoir pressure drops to 400 psig, there will be no operating point for the well with the deepening gas lift system installed. This result is approximately 100 psi lower than the base case, indicating that the well equipped with the retrofit gas lift system will have a slightly extended operational life.

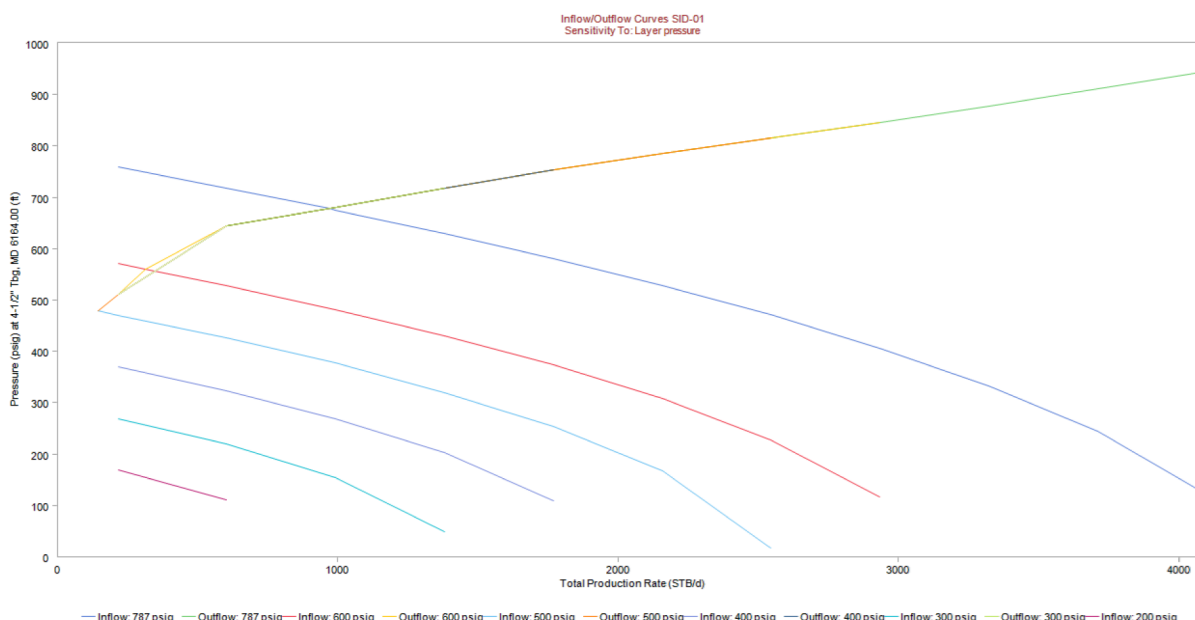


Figure 15—Flow curves retrofit gas lift – Sensitivities on Reservoir Pressure

Table 5—Production summary retrofit gas lift – Sensitivities on Reservoir Pressure

Operating Pressure	Liquid Rate	Oil Rate	Water Rate	Formation Gas Rate	Injection Gas Rate	Water Cut	Produced GOR	Injection MD	Status	Sens: Layer pressure
(psig)	(STB/d)	(STB/d)	(STB/d)	(MMSCF/d)	(MMSCF/d)	(Fraction)	(SCF/STB)	(ft)		(psig)
478.64	143.1	114.3	28.8	0.03	3.1	0.201	27385.18	5466	No operating point	400
560.79	312.8	249.9	62.9	0.07	3.1	0.201	12684.77	5466	Stable	500
677.92	969.5	774.6	194.9	0.22	3.1	0.201	4285.95	5466	Stable	787

Methodology and Installation Procedure

Prior to retrofitting the gas lift, matrix acidizing was performed on the producing zones of the well, SSD#1, SSD#2 and the open hole section to increase the permeability of the perforations. Gas lift valve on the straddled GLM#4 was also changed to the largest available orifice (32/64") to reduce the pressure drop

across the whole gas lift flow path on the retrofit gas lift system. Given the limited space on the platform and the remote area of the operation, a catenary system using an offshore barge was deployed for the coiled tubing unit on the well head platform. Due to these configurations, the length of the BHA that could be accommodated was also limited.



Figure 16—Wellhead Platform

The objective of operation was to deepen the gas lift injection by installing a straddle packer system that directed the lift gas using crossflow sub towards a coiled tubing tailpipe as previously shown in Figure 4. Given the horizontal profile of the well and multiple GLM to be passed to reach the setting depth, tubing clearance was critical. Drift run was performed to confirm access for the gas lift straddle to a depth of 4500 ftMD, around 35 ft below the straddled GLM. An assembly with the largest OD (outside diameter) of 3.66" was used to resemble the OD of the packer.

The first step of the installation was to deploy the coiled tubing tail pipe. The lower BHA consisted of a upward jetting nozzle for directing lift gas upward, a double flapper check valve for pressure control and a grapple connector that were made up at the end of coiled tubing string with details on Figure 17. This lower BHA served as the new gas lift valve and injection point in the retrofit gas lift system. Nodal analysis showed that injection depth of 5466 ftMD, which is 1000 ft deeper than the lowest GLM, was suitable for the operation.

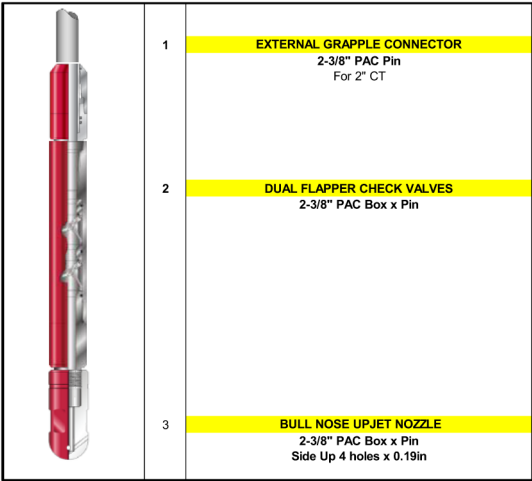


Figure 17—End of tailpipe assembly

979 ft of 2" OD coiled tubing was then run into the hole, that length was required to cover the distance between the straddled GLM to the optimum injection depth modeled previously. Next step is to cut the coiled tubing for it to be hung by the lower packer assembly. Well pressure control and monitoring were critical to ensure the safety of the cutting operation. The BOP pipe-slip rams were then closed, and inflow testing was conducted before disengaging the pressure control lubricator. The reel side of the coiled tubing was opened to inflow test the flapper check valve as well. With this setup, the BOP served as the annular barrier and the double flapper check valves at the end of the tail pipe served as the coiled tubing barrier.

After ensuring that no build up is observed on the CT and riser pressure, the pressure control stack was disconnected on the quick test sub of the coiled tubing riser. The injector head traction was disengaged to allow relative movement between the coiled tubing and the riser. Then, the CT frame was jacked up to expose 3 feet of coiled tubing work window. The coiled tubing was cut with the necessary length required testing on the inverted coiled tubing connector as shown in Figure 18 and as contingency in case of a test failure of the first connector installation.

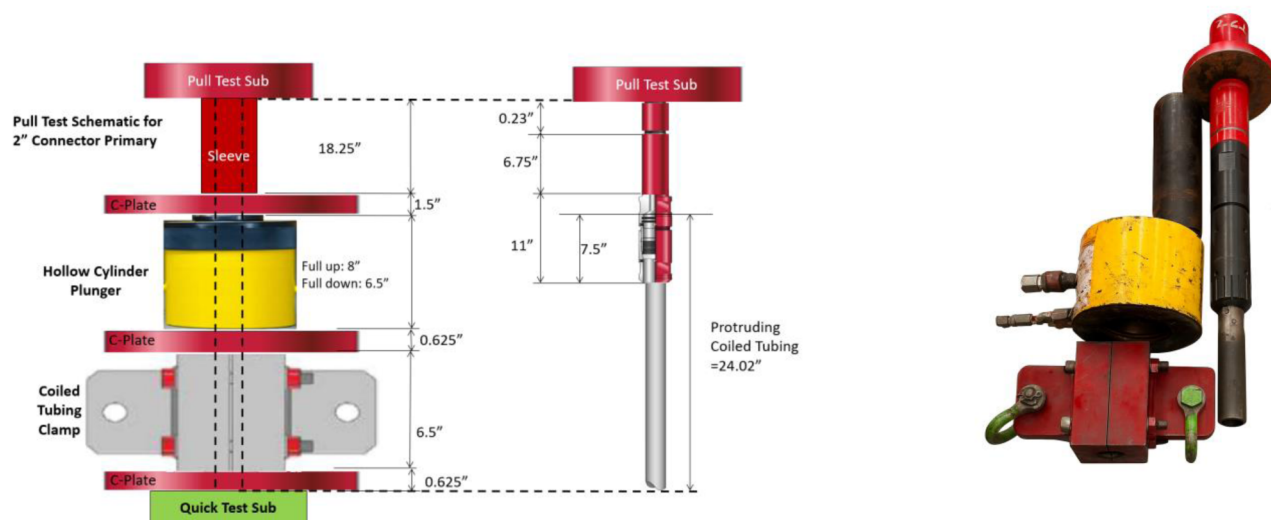


Figure 18—Inverted coiled tubing connector pull test schematic (left), actual simulation (right)

An inverted grapple coiled tubing connector was made up on the protruding CT. The connector integrity is essential in ensuring the effectiveness of the retrofit gas lift system. The inverted position posed a challenge in conducting pull and pressure test on the connector. Pull test were performed using a hollow cylinder jack and pull test plate against a clamp on the CT as shown in Figure 17. The hollow cylinder allowed the jack to be inserted outside of the coiled tubing and pushed the pull test sub so that a pulling force was exerted on the connector. A clamp is made up on the coiled tubing so that the pulling force was not acting against the BOP pipe-slip ram. An appropriate pressure was applied on the hollow cylinder jack so that the cylinder travels upwards to apply 30,000 lbs of tension on the connector. The tension value ensured the connector capability of supporting the coiled tubing weight below.

Following a successful pull test, the testing apparatus was removed to allow pressure testing on the connector. Without the ability to pump from the CT, a special test port was required for inverted coiled tubing connector installation. A test port is integrated into the grapple connector between two sets of O-rings as illustrated in Figure 19, enabling pressure integrity verification in both directions. A pressure test of 5000 psi was applied by a hand pump between the O-rings on the connector.

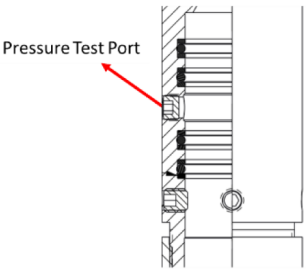


Figure 19—Inverted coiled tubing connector pressure test illustration

With the integrity of the connector confirmed, BHA for setting the lower straddle packer was installed on the injector side of the coiled tubing string. Given the rig-up length constraints of the straddle spacer, precise depth control is essential to correctly position the GLM between the packer elements. With a 15 ft spacer installed, the distance between the packer elements was 23 ft. This required a depth accuracy of ± 11.5 ft for the lower packer installation, which was considered difficult to maintain at a setting depth of 4466 ft. A mechanical profile locator with spring loaded blades was incorporated into the BHA to locate the closest SSD at 5088 ft. This method brought the setting depth closer to the confirmed SSD depth located by the tool, creating a more accurate setting depth.

The lower packer setting BHA consisted of a mechanical profile locator, ball drop circulating sub, a hydraulic setting tool, packer adapter kit, and the lower seal bore retrievable packer which can be seen in Figure 20. Installed below the packer was a slotted joint, which allows production flow from CT annulus to return through the straddle after installation, along with a polished bore receptacle designed to receive the seals from the gas injection tube in the upper packer assembly, isolating the lift gas from the production flow. The packer setting BHA was connected to the inverted connector on the protruding CT by a quick connector which allowed BHA to be made up without rotation.

BHA (schematic)	Item	Description	
	1	EXTERNAL DIMPLE CONNECTOR 2-3/8" PAC Box x Pin For 2" CT	
	2	MOTORHEAD ASSEMBLY 2-3/8" PAC Box x Pin Double Flapper Check Valves Hydraulic Disconnect 7/8" Drop Ball 3 Shear Pins x 900 Psi = 2700 Psi ΔP Circulation Sub 5/8" Drop Ball 3 Shear Pins x 900 Psi = 2700 Psi ΔP Burst Disc : 6000 Psi ΔP	
	3	BALL DROP CIRCULATION SUB 2-3/8" PAC Box x Pin 5/8" Drop Ball	
	4	MECHANICAL PROFILE END LOCATOR 2-3/8" PAC Box x Pin 3.88" OD Profile Dogs	
	5	SWIVEL JOINT 2-3/8" PAC Box x Pin	
	6	CROSSOVER 2-3/8" PAC Box x 2-3/8" EUE Pin	
	6	HYDRAULIC SETTING TOOL 2-3/8" EUE Box x Baker#10 1/2" Ball on Seat 3 Pistons Installed Piston Area = 4.123 in2 / Piston	
	7	366 PACKER ADAPTER KIT Baker #10	
	8	366 LOWER PACKER 3.25" SA - 3G Box Maximum Differential Pressure = 5,000 psi ISO 14310 V0 Validated Setting Force = 44,450 lbs (10 pin x 3,175 lbs) Release Force = 9,600 lbs Setting depth @4477 ft (11 ft below GLM#4)	
	9	SLOTTED PUP JOINT 3.25" SA - 3G Pin x 2.438" SA-2G Box	
	10	POLISHED BORE RECEPTACLE 2.438" SA-2G Pin X 2.375" W TS-8 Pin	
	11	CROSSOVER 2.375" W TS-8 Box x 2-3/8" PAC Pin	
	12	QUICK CONNECTOR 2-3/8" PAC Box x Pin	
	13	CROSSOVER 2-3/8" PAC Box x 1.9" NU Pin	
	14	EXTERNAL GRAPPLE CONNECTOR 1.9" NU Box For 2" CT Equipped with In Situ Pressure Test Port	
		PROTRUDING COILED TUBING	

Figure 20—Lower Packer Setting Bottom Hole Assembly

Once the coiled tubing riser was connected and pressure tested, the BOP pipe/slip ram was disengaged, and the BHA was run in hole to set the packer. Depth correlation was conducted on the closest 3.81" profile, which was at SSD#1 on 5088 ftMD using the mechanical profile locator to ensure accurate setting depth. The work string was then positioned with the packer element at 4477 ft, 11 ft below the straddled GLM. The packer was then set by dropping the ball which landed on the ball drop circulating sub, closing the BHA flow towards the CT annulus. Pressure was then applied on the work string to activate the hydraulic setting tool until the required pack-off force was achieved on the packer elements and anchor. The packer adapter kit was then sheared off indicated by drop in pressure completing the setting sequence of the packer. The packer anchor was then tested by setting down 2000 lbs of weight on top of the packer. Upon confirmation of successful set, the coiled tubing and BHA were returned to surface.

The last step of the installation was to set the upper packer to complete the straddle packer assembly above the GLM. Setting the upper packer assembly utilized the same tool string configuration as the lower packer run as shown in Figure 21, apart from the mechanical profile locator, which was not included due to the depth has been determined by the lower packer. Below the packer was a 15 ft spacer pup joint, anchor seal assembly for latching and sealing into the seal bore of the lower packer, crossflow sub and gas injection stinger. The crossflow sub directs the lift gas from the GLM into the straddle straight down to the gas injection pipe and allows production flow to pass through the straddle assembly up towards the tubing as shown in Figure 3. The gas injection pipe has a stinger which landed on the polished bore receptacle, isolating the flow between lift gas and production flow, and continues the lift gas towards the coiled tubing tailpipe.

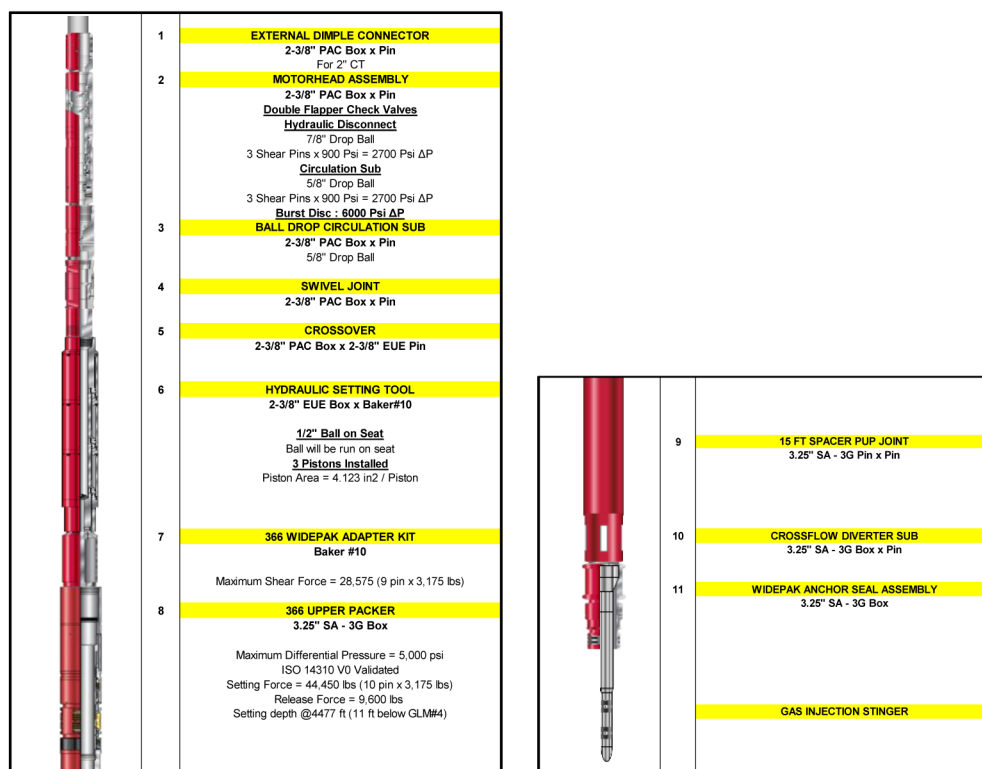


Figure 21—Upper Packer Setting Bottom Hole Assembly

The BHA was run-in hole until the anchor seal latched into the seal bore of the lower packer. Tension was applied to ensure proper latch, setting ball was dropped and pressure was applied into the coiled tubing to activate the hydraulic setting tool. A pressure drop indicated the appropriate pack-off force was already applied to the packer elements and anchor which was determined by the shear pins on the packer adapter

kit. The setting assembly was retrieved to surface upon the retrofit deepening gas lift system installation was completed.

Results

Upon the complete installation of the deepening gas lift system, the well was then shut in to allow pressure build up necessary to kick start the well. With current retrofitted well condition, the lift gas was able to be injected into the casing annulus and enter the tubing through the GLM. Lift gas flows into a port contained by the straddle and was directed down an inner coiled tubing string suspended by the lower packer 1000 ft below the original GLM as shown in Figure 23. By allowing the gas lift to enter deep and closer to the horizontal section of the well, fluid was able to be unloaded more effectively by reducing the hydrostatic pressure across the producing zones.

The well choke was then opened and connected to the well test unit with lift gas injected at 3 MMSCFD. Data was then gathered when the well was producing in stable manner. An immediate increase in well performance was observed. As shown in Table 6, 200% the amount of oil and liquid was obtained compared to previous maximum well capability. These results aligned with the nodal analysis model, showing differences of only 1.49% in oil flow rate and 21.09% in liquid flow rate, respectively. The error resulted from the difficulty in aligning the predictive nodal plot to intersect at the exact production rate observed in the actual well, due to the approach of various multiphase-flow tubing or flowline correlations (Persad, 2005). In this case the error value was considered acceptable for the application. Figure 22 shows the production chart for around 1 year that compares before and after installation of the sytem. While the well was capable of producing at the maximum potential rate, the choke was adjusted to deliver around 500–700 BLPD to maintain sustainable production and still meeting the target. This represented an increase of approximately 200 BLPD throughout the year.

Table 6—Well performance and production results before and after intervention

Condition	Injection MD	Injection Gas Rate	Tubing Head Pressure	Liquid Rate	Oil Rate	Water Rate	Remarks
	(ft)	(MMSCF/d)	(psig)	(STB/d)	(STB/d)	(STB/d)	
Before intervention	4466	3	266.8	561.3	364.84	196.46	
Nodal Analysis result – Deepening Gas Lift System	5466	3.1	261.3	964.1	770.3	193.8	Oil Rate = 211% of initial condition Liquid Rate = 171% of initial condition
Well test data after installation	5466	3	274.1	1167.4	758.81	408.59	Oil Rate = 207% of initial condition Liquid Rate = 208% of initial condition
Difference between model and actual				21.09 %	1.49 %		

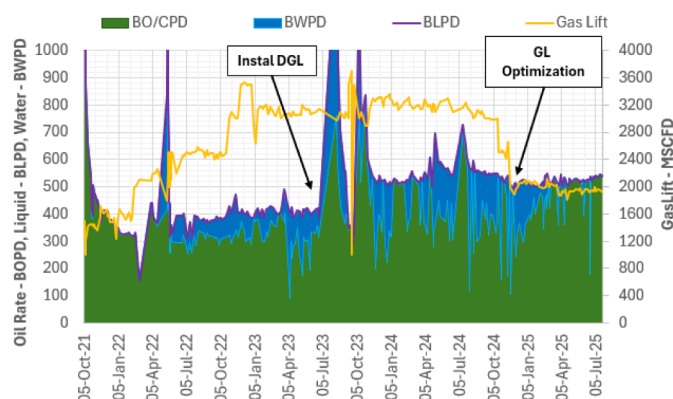


Figure 22—Well S1 production graph after intervention

Installation of the retrofit deepening gas lift system took 73 hours of operations including all the 3 runs required, which was 40% of the overall campaign on the well. The live well operation meant that no additional preparation was required to shut in the well, and minimal downtime was needed to bring it back online, achieving the intended objective in terms of intervention effectiveness.

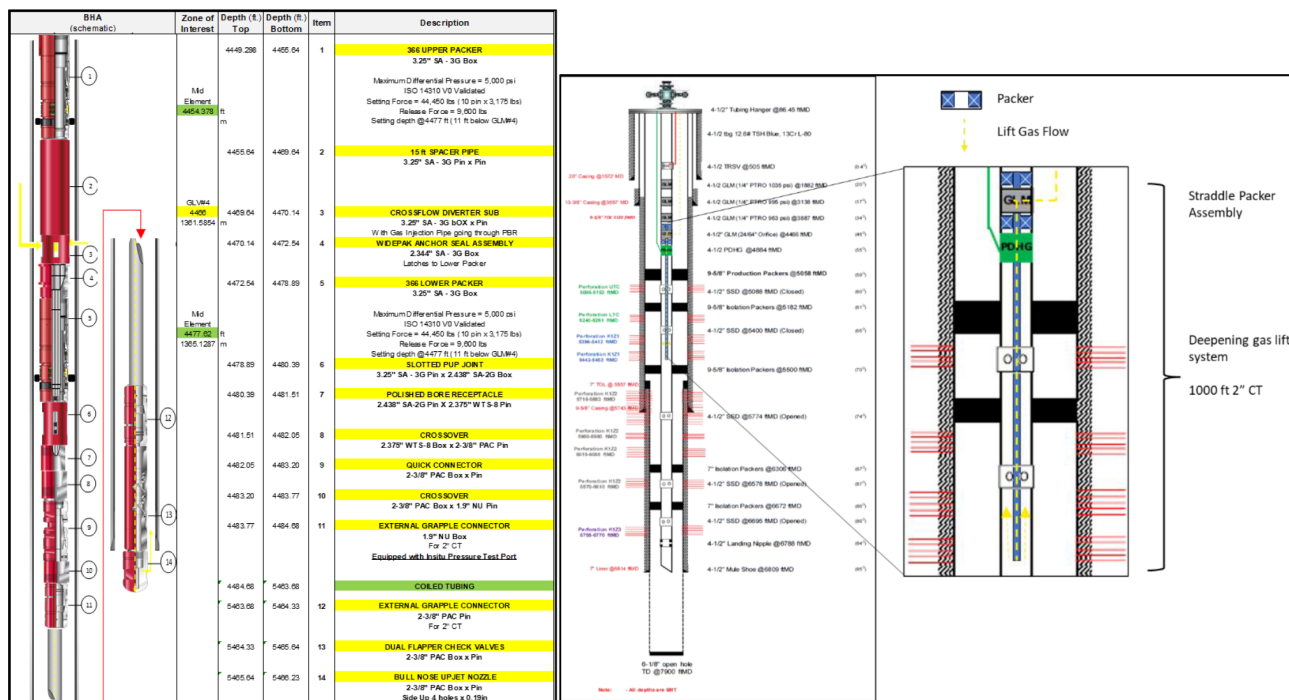


Figure 23—Deepening gas lift system installed in the well

Conclusions

The thru tubing retrofit deepening gas lift system was successfully deployed and proved effective in enhancing gas injection depth within the well. By positioning the gas lift valve deeper in the wellbore, the system allowed for improved unloading of accumulated liquids and better reservoir energy utilization. The operation demonstrated the importance of thorough job planning, with a strong emphasis on gas lift optimization, operational feasibility, and cost-effectiveness—successfully overcoming the limitations encountered on location.

Gas lift optimization was successfully achieved by Nodal Analysis. The close alignment between measured and modeled results demonstrates that nodal analysis modeling is a valuable tool for optimizing system installation and performance prediction, providing reliable estimates that support data driven decision making in well intervention planning.

This operation highlights a significant and novel technology deployment in the area. It serves as a viable solution for similar underperforming or aging assets experiencing declining reservoir pressure and liquid loading beyond the existing gas lift injection depth.

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